

# A Novel Method for Trust-Sensitive Design: Applying Conjoint Analysis to Machine-Assisted Genome Annotation

Melanie McGrath

Data61

CSIRO

Melbourne, Australia

The University of Melbourne

Parkville, Victoria, Australia

melaniejoymcgrath@gmail.com

Patrick Cooper

Data61

CSIRO

Clayton, Victoria, Australia

patrick.cooper@data61.csiro.au

Andreas Duenser\*

Data61

CSIRO

Hobart, Tasmania, Australia

andreas.duenser@csiro.au

Jessica Irons

Data61

CSIRO

Eveleigh, NSW, Australia

jessica.irons@data61.csiro.au

Xiaomei Li

Agriculture and Food

CSIRO

Marsfield, NSW, Australia

maisie.zhang@csiro.au

Rad Suchecki

Agriculture and Food

CSIRO

Adelaide, SA, Australia

r.suchecki@gmail.com

## Abstract

Human trust is recognized as being central to integration of AI systems into organizational settings and workflows. However, trust is a highly context-dependent, state-based attitude that can be influenced by hundreds of user, technology, and environmental factors. In this paper we describe the novel application of conjoint analysis, an established behavioral science method, to quantify the relative contribution of four design features to trust in an AI application to support scientists engaged in genome annotation. We found that Transparency had the greatest relative importance to trust, followed by Confidence, Control, and Accuracy. Our results demonstrate the utility of a conjoint analysis approach for identifying and prioritizing trust-sensitive design features, offering a cost- and time-effective approach to guide development efforts.

## CCS Concepts

• Human-centred computing; • Human computer interaction; • Empirical studies in HCI;

## Keywords

trust, artificial intelligence, genome annotation, conjoint analysis

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\*Corresponding author

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## 1 Introduction

Trust is critical to the safe and effective use of artificial intelligence (AI) – when we don't know how a system works, it's often our level of trust that determines how willing we are to use it. Consequently, researchers and practitioners need to understand the features that influence trust in specific AI applications and the context in which they are used. These may be features of the human user, the technology itself, and the task or task environment [12]. However, identifying the 'ingredients' for trust in any given application is complicated by the fact that more than 470 individual factors have been shown to influence trust in AI [15]. Further complicating matters, trust is highly context-dependent, and each AI application requires its own model for trust [14]. Currently researchers and practitioners need to comb an ever-expanding and largely unstructured scientific literature to determine which of the many factors influencing trust will carry the greater weight in their specific application.

Designers and developers of AI applications are likely to be most interested in the characteristics of the system itself. In most cases developers will have little control over the task or the characteristics of the people using the AI application, but they can make decisions regarding the design of the application. In this paper we describe the novel application of conjoint analysis, an established behavioral science method, to quantify the relative contribution of a range of design factors to trust in an AI application being designed to support scientists engaged in genome annotation. This process provides empirical data on end-user preferences and priorities and thus directly assists designers and developers in making well informed decisions.

### 1.1 GeneWhisperer: An application for machine-assisted genome annotation

Genome annotation is essential for uncovering functional elements within genomes. While manual curation remains a key approach for identifying gene functions, especially those overlooked by automated annotation pipelines, it is often a time-intensive and demanding process [4]. Although various software tools have been

developed to assist human experts with annotation tasks, they lack integration and streamlining, which hinders the efficiency of manual curation. As a result, human annotators frequently need to switch contexts and manually transfer data between disparate systems.

An interdisciplinary team of researchers at Australia’s Commonwealth Scientific and Industrial Research Organisation (CSIRO) has developed GeneWhisperer, an innovative platform that combines domain-specific tools with human-AI collaboration through a language-based interface. GeneWhisperer features a user-friendly conversational interface that allows biologists and biocurators to input natural language queries and receive corresponding results. In the backend of the platform, a custom AI agent leverages prompt-tuned LLMs to extract and synthesize information from comprehensive databases, scientific literature, knowledge graphs, and online sources. The AI agent functions as an intelligent entity that mimics human cognitive abilities, allowing it to perceive its environment, plan, and take actions to achieve specific goals [21]. AI agents have demonstrated significant potential in accelerating knowledge discovery across multiple domains [20] and in reducing the manual effort required for data retrieval and integration tasks [7].

## 1.2 Conjoint analysis

As a tool designed to support scientists in their work, it is vital that these users trust GeneWhisperer sufficiently to incorporate it into their workflows [11]. Conjoint analysis is a methodological approach with the potential to identify not only the factors likely to influence trust in GeneWhisperer, but also to quantify their relative importance. Conjoint analysis is a well-established method in consumer psychology and economics and has traditionally been used to quantify people’s preferences for a particular combination of attributes in a product or service [6, 8]. In this experimental paradigm the features of a particular product or service are systematically varied to create depictions of a number of alternative products termed “concepts”. A sample of end-users is then presented with a series of comparisons of these concepts and asked to choose between them on a given parameter. In our study a sample of intended future end-users of GeneWhisperer were asked to compare the concepts on the basis of which they would trust more, drawing on their domain knowledge of genome annotation. The amount of value each feature adds or subtracts is calculated from user responses. The goal was to provide guidance for ongoing GeneWhisperer design decisions.

**1.2.1 Attributes.** Within a conjoint analysis the features that are systematically varied are referred to as ‘attributes’. These represent the design factors believed to influence trust in the AI application [1]. Through a process described in the Method section, the following four attributes were identified for testing in our analysis of trust in GeneWhisperer: Transparency, Control, Accuracy, and Confidence. Transparency refers to how GeneWhisperer shares information about the backend processes it uses to select tools and process results. Control refers to whether the user is able to control GeneWhisperer’s choice of tools and how it uses them. Accuracy refers to the performance of the system in terms of how frequently it gives an appropriate response to the query. Finally, Confidence

refers to whether GeneWhisperer communicates how confident it is in the accuracy of the results it is providing.

**1.2.2 Levels.** ‘Levels’ represent the design options within each attribute [1]. For example, within the Control attribute the specified levels were *full user control* (the user manually selects the tools and how they are used), *full autonomy* (GeneWhisperer selects the tools and executes the process automatically), and *hybrid control* (users can intervene and exert control as needed). A full list of levels for each attribute are given in Table 1. The range of concepts presented to our end-users represent combinations of these levels across attributes. Through a series of comparisons end-users effectively indicate their priorities when it comes to developing trust in GeneWhisperer.

## 2 Method

All studies were approved by the CSIRO Social Science Human Research Ethics Committee and were conducted in accordance with local legislation and institutional requirements.

### 2.1 Selecting attributes and defining levels

Attributes are the factors expected to influence end-users’ trust in a given AI application. Given that conjoint analysis in this setting is intended to inform the design of GeneWhisperer, selected attributes needed to represent a balance between what is likely to be important to end-users and what is relevant to system designers.

An initial set of attributes were identified via semi-structured interviews with 12 scientists engaged in genome annotation (for more details on methodology see [11]). These interviews included questions regarding user characteristics, task context, scientists’ attitudes towards the technology currently used in their role, their attitudes and expectations regarding future incorporation of AI into their workflows, and direct questions about trust in AI in this work context [11]. Thematic analysis of interview transcripts revealed factors likely to be relevant to trust in GeneWhisperer.

The first key theme was that of transparency. Multiple interviewees expressed that before they could rely on an AI system’s output, they would want to know what decisions it was making in the background – *“I would want it to be quite specific about when it’s removing things”*. A second theme that emerged was regarding control over the results, that is, not simply knowing what choices an AI system was making, but being the one to make those choices – *“I want to be the one deciding on when things get removed”*. Interviewees noted the importance of the accuracy of the results. AI system accuracy can be characterized in many ways. It might refer to the absolute correctness of a result compared to a reference standard, or to the relevance of a set of outcomes. For our purposes, it was important that the accuracy attribute align with the context and priorities of end-users, and provide meaningful information for system designers. In genome annotation, since ground truth is often unavailable or unclear, using an absolute correctness measure was not suitable. In our interviews scientists referred to the importance of relevance of the results to their query, with some suggestion of a preference for results to be overinclusive and tend towards false positives, rather than tending towards false negatives in which relevant results may be missed – *“There’s false negatives ... you missed out on something that could potentially be evidence*

**Table 1: Attributes and levels**

| Attributes   | Level 1                                                                                                                                                               | Level 2                                                                                                                                                        | Level 3                                                                                                                               |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Transparency | <i>Full transparency:</i> GeneWhisperer shares everything, including the tools chosen, intermediate processing results, data sources used, and detailed explanations. | <i>Selective transparency:</i> Upon request, GeneWhisperer provides details on specific aspects, for example, the chosen tool, the data used, or explanations. | <i>Limited transparency:</i> Only the final results are presented, with not access to the underlying process.                         |
| Control      | <i>Full user control:</i> You have maximum control by manually selecting the tools and how they are used.                                                             | <i>Full autonomy:</i> After receiving your input, GeneWhisperer selects the tools and executes the process automatically without further interaction.          | <i>Hybrid control:</i> GeneWhisperer operates with pre-defined default settings, allowing you to intervene and modify them as needed. |
| Accuracy     | <i>More false negatives:</i> GeneWhisperer is more specific and might miss relevant information (false negatives).                                                    | <i>More false positives:</i> GeneWhisperer is more sensitive and might include irrelevant or incorrect information (false positives).                          |                                                                                                                                       |
| Confidence   | <i>Always:</i> GeneWhisperer always expresses its confidence level alongside results.                                                                                 | <i>On demand:</i> GeneWhisperer doesn't report confidence levels, but you can request them.                                                                    | <i>None:</i> GeneWhisperer presents the results without any indication of confidence.                                                 |

for how these evolved . . . As for false positives, that's more just you're publishing something that's not actually true . . . and maybe it gives you some spurious signal that you overinterpret". Informed by these interviews and consultation with domain experts in the project team, the accuracy attribute sought to test this preference. Domain experts in the project team also proposed the inclusion of confidence as an attribute, that is, whether the system provides confidence estimates alongside its results. This provided the final set of four attributes, which were then defined in clear and accessible terminology for presentation to end-users. Once attributes were selected, the project team identified levels within each attribute that represent meaningful design variations (see Table 1).

## 2.2 Participants and procedure

Participants were recruited from among genomics scientists in the Genomics Community of Practice at the CSIRO and the Australian BioCommons Genomics group. Interested scientists were directed to the Conjointly survey platform (<https://conjointly.com/>) where they read a Participant Information Statement and indicated their consent to participate. Participants provided information regarding the types of genome annotation they perform in their current and previous roles and how many years of experience they have in genome annotation. The final sample of participants was 17 scientists with, on average, eight years of experience in the field.

These participants were shown a 1:55 min video explaining GeneWhisperer and were introduced to each attribute and its definition. After the attributes had been defined, a comparison between concepts was presented, with participants asked to choose "Which of the following versions of GeneWhisperer would you trust more to assist you in your work?" (see Figure 1 for an example). Twelve of these comparisons were presented in a fractional factorial choice

design optimized by the survey platform for balance and overlap [1].

## 3 Results

A conjoint analysis quantifies the relative importance of attributes and levels to user trust in the target product using a metric called partworth utility (PU), that reflects the relative value participants assign to different attribute levels based on their preferences. In the present study PUs were calculated using Markov Chain Monte Carlo Hierarchical Bayes estimation. It is important to note that all PUs reported here are relative only to the attributes and levels specified in the current study. These values are not absolute and cannot be compared with analyses featuring different attributes.

### 3.1 Relative importance of attributes

The relative importance of attributes is calculated by first determining the range of preference within each attribute for each participant. An importance ratio of each attribute for each participant is then calculated by dividing the preference range for a specific attribute by the sum of preference ranges for all attributes. The relative importance of each attribute is the average importance ratio across all participants. Attribute utilities for GeneWhisperer are given in Figure 2.

Transparency was found to be of greatest importance to projected levels of trust in GeneWhisperer and was twice as important to our participants as Accuracy (conceptualized as a tendency towards false negatives or false positives).

### 3.2 Relative importance of levels

We further explored what level of each attribute the average scientist in our sample would trust more. Values depicted in Figure 3

Which of the following versions of GeneWhisperer would you trust more to assist you in your work?

|              | Version 1                                                                                                                                               | Version 2                                                                                                                                      | Version 3                                                                                                                                                      |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Transparency | Selective transparency: Upon request, GeneWhisperer provides details on specific aspects, for example, the chosen tool, the data used, or explanations. | Limited transparency: Only the final results are presented, with not access to the underlying process.                                         | Full transparency: GeneWhisperer shares everything, including the tools chosen, intermediate processing results, data sources used, and detailed explanations. |
| Control      | Full user control: You have maximum control by manually selecting the tools and how they are used.                                                      | Full autonomy: After receiving your input, GeneWhisperer selects the tools and executes the process automatically without further interaction. | Hybrid control: GeneWhisperer operates with pre-defined default settings, allowing you to intervene and modify them as needed.                                 |
| Accuracy     | More false positives: GeneWhisperer is more sensitive and might include irrelevant or incorrect information (false positives).                          | More false positives: GeneWhisperer is more sensitive and might include irrelevant or incorrect information (false positives).                 | More false negatives: GeneWhisperer is more specific and might miss relevant information (false negatives).                                                    |
| Confidence   | Always: GeneWhisperer always expresses its confidence level alongside results.                                                                          | On demand: GeneWhisperer doesn't report confidence levels, but you can request them.                                                           | None: GeneWhisperer presents the results without any indication of confidence.                                                                                 |

Figure 1: An example of the comparison between concepts presented to participants

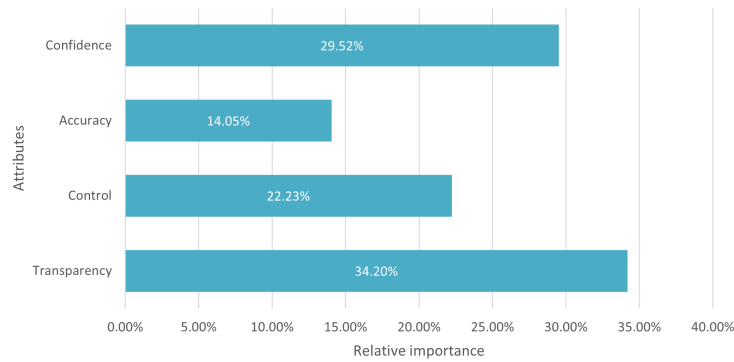


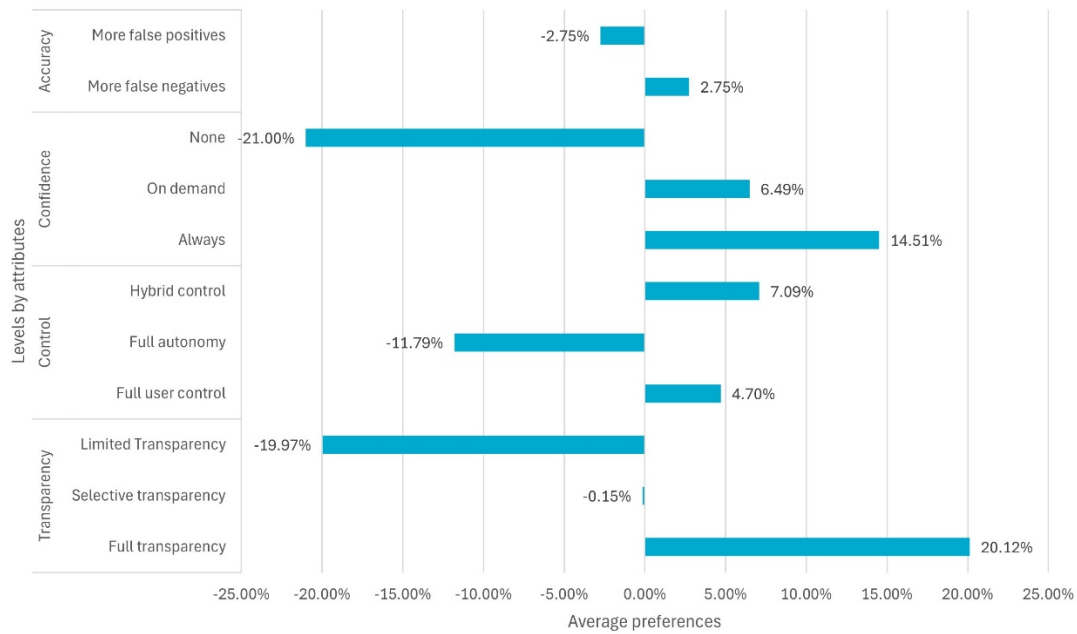
Figure 2: Relative importance of attributes for trust in GeneWhisperer

are 0-centered within attribute, with high positive values equating to relatively high levels of trust associated with this feature. Values are calculated based on the relative importance of the level against other levels within the same attribute,

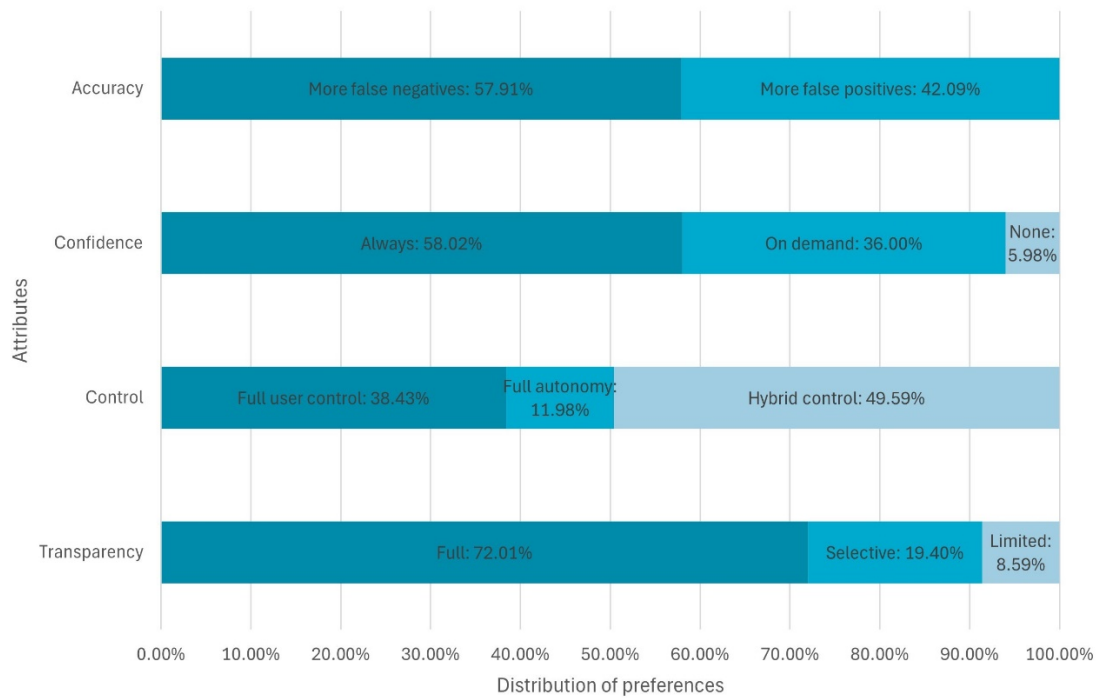
The attribute PUs previously reported indicated that Transparency was the most consequential attribute in determining trust in GeneWhisperer. The level PUs displayed in Figure 3 clarify that within this attribute, the average preference of our participants was for full transparency. Versions of GeneWhisperer with either limited or selective transparency were trusted less. Confidence was the next most important attribute, and here we saw that confidence information provided either by default (Always) or on demand contributed to higher reported trust. Versions of GeneWhisperer that did not provide any confidence information were trusted significantly less. We saw a similar pattern of preferences for levels

within the Control attribute. Both full user control and hybrid control were advantageous to trust, with participants considerably less likely to trust a version of the AI platform that operated in a fully autonomous fashion. Finally, analysis of Accuracy showed minimal difference in preferences for a tendency towards more specificity (with a greater risk of false negatives) and more sensitivity (with greater risk of false positives).

These results give us the average preference of the scientists in our sample. We can also inspect the distribution of preferences across our participants in Figure 4.



**Figure 3: Relative importance of levels for trust in GeneWhisperer**



**Figure 4: Distribution of level preferences across participants**

## 4 Discussion

Scientists engaged in genome annotation were presented with four potential attributes of the GeneWhisper platform for machine-assisted genome annotation. Participants indicated that Transparency, followed by communicating Confidence levels, was most likely to influence their trust in the platform. Specifically, full transparency regarding the choices being made by the system led to greater trust in 72.01% of participants. 58.02% of surveyed scientists preferred that GeneWhisperer always communicate a level of confidence in the accuracy of the results being provided. However, a further 36.00% reported that providing confidence information on demand would increase their trust, suggesting that providing no confidence information would limit trust development.

Level of Control over GeneWhisperer's choice of tools, and Accuracy preferences towards false positives or false negatives, were relatively less important in forming user trust in the system than Transparency and communicating Confidence. This is consistent with previous work indicating that factors associated with communication and information exchange are key to trust in an AI application [5, 16, 19, 22].

Our application of conjoint analysis to the design of GeneWhisperer illustrates the utility of this methodological approach for identifying and predicting the factors most relevant to user trust for a given application. This has the potential to be of great benefit to designers and developers. Human trust is recognized as being central to integration of AI systems into organizational settings and workflows [10, 13]. However, trust is a highly context-dependent, state-based attitude that can be influenced by hundreds of user, technology, and environmental factors [12, 15]. Conjoint analysis permits designers to quantify the relative importance of a range of trust-sensitive design features before committing to expensive and time-consuming prototype development. It can further inform the design process through application to competing options for operationalizing preferred levels. For example, in deciding how to deliver the full transparency predicted to facilitate trust in GeneWhisperer, designers may conduct a secondary conjoint analysis comparing types of user interfaces or communication modes. A further benefit is that, unlike testing via prototypes, the levels in a conjoint analysis aren't limited to current capability. This approach can be used to provide a sense of what future capability or functionality is worth an investment of time and effort. For example, a team may decide not to spend months building a transparency module into a new AI system if a conjoint analysis indicates the end result will make little difference to user trust.

## 5 CONCLUSION and future work

In this paper, we described the novel application of an economic and behavioral science analytic method to quantifying the factors influencing trust in an AI product. In traditional applications of conjoint analysis to market research or evaluation of attitudes, the results have not commonly been experimentally validated. In market research, findings are frequently verified against market share of the final product [2]. In the evaluation of attitudes, such as the effect of candidates' gender on voter choice, validation is often conducted in the form of benchmarking against existing datasets [9]. Our next steps in the GeneWhisperer project will be to conduct

a behavioral validation of our findings as part of prototype user-testing. Specifically, we will examine whether inclusion of the features predicted to influence trust is reflected in reports of trust and trust behavior in interaction with prototypes.

While significant, trust is not the only factor that influences a user's willingness to rely on an AI system. Numerous other influences on technology acceptance have been identified in an extensive literature [3, 17]. These include perceived usefulness, perceived ease-of-use, and social norms [18]. The present study investigates trust as the outcome variable of interest, however, future work will investigate the relative weight of trust against other such factors for a full understanding of features likely to contribute to adoption of GeneWhisperer.

Informed by comprehensive, semi-structured interviews, we identified four attributes expected to affect trust in an AI prototype supporting scientists in genome annotation. Using conjoint analysis, we found that of these four attributes, Transparency had the greatest relative importance to trust, followed by Confidence, Control, and Accuracy. These findings underscore the importance of information exchange in designing AI systems that effectively integrate into human-centric workflows. We further demonstrated how conjoint analysis can be used to quantify which levels of each of these attributes builds user trust. Our work demonstrates the utility of a conjoint analysis approach for identifying and prioritizing trust-sensitive design features, offering a cost- and time-effective approach to guide development efforts.

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